Denoising Diffusion Implicit Models (DDIMs)

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Abstract

Denoising Diffusion Probabilistic Models (DDPMs) generate high-quality samples but are slow, requiring up to $T \approx 1000$ sequential denoising steps. Song et al. [2020a] introduce *Denoising Diffusion Implicit Models (DDIMs)*, which reuse a pretrained DDPM but produce samples in $S \ll T$ steps, with the option of fully deterministic sampling. The key idea is that the standard DDPM training loss depends only on the *marginals* $q(\boldsymbol{x}_t \mid \boldsymbol{x}_0)$, not on the joint $q(\boldsymbol{x}_{1:T} \mid \boldsymbol{x}_0)$. This allows one to replace the Markovian forward process by a non-Markovian family q_σ with the same marginals, derive a corresponding parametric reverse process, and prove that its variational objective reduces to the DDPM's objective.

1 Introduction

Given samples from an unknown data distribution $q(\boldsymbol{x}_0)$ on $\mathbb{R}^d$, we seek a model $p_\theta(\boldsymbol{x}_0)$ that approximates $q(\boldsymbol{x}_0)$ and is easy to sample from. DDPMs [Ho et al., 2020] achieve this by defining T latent variables $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_T \in \mathbb{R}^d$ and a fixed forward (noising) Markov chain $q(\boldsymbol{x}_{1:T} \mid \boldsymbol{x}_0)$, then learning a parametric reverse Markov chain $p_\theta(\boldsymbol{x}_{0:T})$ that progressively denoises Gaussian noise into a sample.

DDPM sample quality is competitive with GANs, but generation is expensive: each sample requires T sequential network evaluations. Ho et al. [2020] use $T = 1000$, which on a single Nvidia 2080 Ti translates to roughly 20 hours for 50K 32×32 CIFAR-10 images, and around 1000 hours for 50K 256×256 images. GANs by contrast generate the same batch of CIFAR-10 images in under a minute.

The goal of Song et al. [2020a] is to sample in $S \ll T$ steps using the *same* pretrained noise-prediction network $\boldsymbol{\epsilon}_\theta^{(t)}$, by changing the inference procedure without changing the trained weights. Their starting point is the observation that the DDPM denoising objective L_1 (Section 3.1) is a sum, over time indices t, of terms that depend only on the per-step marginals $q(\boldsymbol{x}_t \mid \boldsymbol{x}_0)$ and not on the full joint $q(\boldsymbol{x}_{1:T} \mid \boldsymbol{x}_0)$. Since infinitely many joint distributions share a given set of marginals, one might be free to replace the Markovian forward process of DDPM with any other joint that has the same marginals, leaving the training objective unaffected. Song et al. [2020a] build a parametric family of such non-Markovian inference processes $\{q_\sigma\}_{\boldsymbol{\sigma} \in \mathbb{R}^T_{\geq 0}}$. The choice $\boldsymbol{\sigma} = \mathbf{0}$ collapses the corresponding generative process to a deterministic map, which is the DDIM. A related construction over a sub-sequence of time indices then yields the sampling speedup.

We follow throughout the notation of Song et al. [2020a]. The clean data is denoted $\boldsymbol{x}_0 \in \mathbb{R}^d$, with $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_T$ a sequence of latent variables in the same ambient space. The forward process q is fixed and non-parametric, while the reverse process p_θ is parametric and trained. The schedule $\alpha_{1:T} \in (0, 1]^T$ is decreasing; we use α_t for what Ho et al. [2020] call $\bar{\alpha}_t$ in the original DDPM paper. The map $\boldsymbol{\epsilon}_\theta^{(t)} : \mathbb{R}^d \to \mathbb{R}^d$ is the noise-prediction network at step t.

38th Conference on Neural Information Processing Systems (NeurIPS 2024).

2 Related Work

Diffusion models trace back to Sohl-Dickstein et al. [2015], who introduced a variational training procedure for slowly destroying and reconstructing data through a Markov chain. Ho et al. [2020] brought the framework to image generation by adopting Gaussian transitions, a noise-prediction parametrisation, and the simplified MSE loss L_1, achieving FID competitive with GANs on CIFAR-10. The DDPM construction is the starting point of the present work, and we recall its details in Section 3.1.

A second line of work casts the same forward process as the discretisation of a stochastic differential equation. Song et al. [2020b] unify denoising-diffusion and score-matching under continuous-time SDEs and introduce the variance-exploding (VE) and variance-preserving (VP) families together with their associated probability-flow ODEs. The DDIM generative process turns out to coincide with an Euler discretisation of the VE probability-flow ODE in the limit, which connects the present work to score-based generative modelling. We will not be talking about this interpretation of DDIMs in this report.

The objective derived for DDPMs is structurally a variational lower bound and resembles the ELBO of a VAE [Kingma and Welling, 2013], with the latent set taken as $\boldsymbol{x}_{1:T}$ and a fixed inference distribution. This perspective motivates the variational treatment in Section 3.4, where the same identity is used with a non-Markovian inference family q_σ.

Several other approaches to faster diffusion sampling have been proposed concurrently or subsequently. The route taken by Song et al. [2020a], and followed here, is distinct in that it requires no additional training and reuses an existing DDPM without modification.

3 Methodology

3.1 Preliminaries: DDPMs

We recall the DDPM construction in the form we need later. The forward (inference) process is a fixed Markov chain that progressively adds Gaussian noise to the data

$$q(\boldsymbol{x}_{1:T} \mid \boldsymbol{x}_0) = \prod_{t=1}^{T} q(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}), \qquad q(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}) = \mathcal{N}\Big(\sqrt{\tfrac{\alpha_t}{\alpha_{t-1}}}\,\boldsymbol{x}_{t-1},\ \big(1 - \tfrac{\alpha_t}{\alpha_{t-1}}\big)\boldsymbol{I}\Big) \tag{1}$$

and a short calculation shows that this chain admits a closed-form marginal

$$q(\boldsymbol{x}_t \mid \boldsymbol{x}_0) = \mathcal{N}(\sqrt{\alpha_t}\,\boldsymbol{x}_0,\ (1-\alpha_t)\boldsymbol{I}) \iff \boldsymbol{x}_t = \sqrt{\alpha_t}\,\boldsymbol{x}_0 + \sqrt{1-\alpha_t}\,\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}) \tag{2}$$

The schedule is chosen so that $\alpha_T \approx 0$, in which case $q(\boldsymbol{x}_T \mid \boldsymbol{x}_0)$ is close to a standard Gaussian for every $\boldsymbol{x}_0$. This justifies setting the prior of the generative process to $p_\theta(\boldsymbol{x}_T) = \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$.

The generative (reverse) process is a parametric Markov chain running in reverse

$$p_\theta(\boldsymbol{x}_{0:T}) = p_\theta(\boldsymbol{x}_T) \prod_{t=1}^{T} p_\theta^{(t)}(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t), \qquad p_\theta^{(t)}(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t) = \mathcal{N}\big(\mu_\theta^{(t)}(\boldsymbol{x}_t),\ \sigma_t^2 \boldsymbol{I}\big) \tag{3}$$

and is trained by maximising the standard variational lower bound on $\log p_\theta(\boldsymbol{x}_0)$

$$\log p_\theta(\boldsymbol{x}_0) \ \geq\ \mathbb{E}_{q(\boldsymbol{x}_{1:T} \mid \boldsymbol{x}_0)}\big[\log p_\theta(\boldsymbol{x}_{0:T}) - \log q(\boldsymbol{x}_{1:T} \mid \boldsymbol{x}_0)\big] \tag{4}$$

This ELBO has the same structure as in a variational autoencoder, with two differences: the inference distribution q is fixed and non-parametric rather than a learned encoder, and the latent space has the same dimension as the data at every time step rather than passing through a low-dimensional bottleneck.

Under the Gaussian assumptions above, the bound simplifies considerably. Standard manipulations [Ho et al., 2020] rewrite (4) as a sum of KL divergences between Gaussians

$$\begin{aligned} \mathbb{E}_{q(\boldsymbol{x}_0)}\big[-\log p_\theta(\boldsymbol{x}_0)\big] \ \leq\ \mathbb{E}_q\Bigg[& \underbrace{D_{\mathrm{KL}}\big(q(\boldsymbol{x}_T \mid \boldsymbol{x}_0) \,\big\|\, p_\theta(\boldsymbol{x}_T)\big)}_{L_T} \ \underbrace{-\ \log p_\theta^{(1)}(\boldsymbol{x}_0 \mid \boldsymbol{x}_1)}_{L_0} \\ & + \sum_{t=2}^{T} \underbrace{D_{\mathrm{KL}}\big(q(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0) \,\big\|\, p_\theta^{(t)}(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t)\big)}_{L_{t-1}} \Bigg], \end{aligned} \tag{5}$$

where L_T does not depend on θ (since $q(\boldsymbol{x}_T \mid \boldsymbol{x}_0)$ is fixed and $p_\theta(\boldsymbol{x}_T) = \mathcal{N}(\mathbf{0}, \boldsymbol{I})$). The true posterior $q(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0) = \mathcal{N}\big(\tilde{\mu}_t(\boldsymbol{x}_t, \boldsymbol{x}_0),\, \tilde{\beta}_t \boldsymbol{I}\big)$ has closed-form mean

$$\tilde{\mu}_t(\boldsymbol{x}_t, \boldsymbol{x}_0) \;=\; \frac{\sqrt{\alpha_{t-1}}\,\beta_t}{1-\alpha_t}\,\boldsymbol{x}_0 \;+\; \frac{\sqrt{\alpha_t/\alpha_{t-1}}\,(1-\alpha_{t-1})}{1-\alpha_t}\,\boldsymbol{x}_t \tag{6}$$

with per-step variance $\beta_t := 1 - \alpha_t/\alpha_{t-1}$ and posterior variance

$$\tilde{\beta}_t \;:=\; \frac{1-\alpha_{t-1}}{1-\alpha_t}\,\beta_t \;=\; \frac{1-\alpha_{t-1}}{1-\alpha_t}\big(1-\alpha_t/\alpha_{t-1}\big) \tag{7}$$

Since $\boldsymbol{x}_t = \sqrt{\alpha_t}\,\boldsymbol{x}_0 + \sqrt{1-\alpha_t}\,\boldsymbol{\epsilon}$ is invertible in $\boldsymbol{x}_0$, we substitute $\boldsymbol{x}_0 = (\boldsymbol{x}_t - \sqrt{1-\alpha_t}\,\boldsymbol{\epsilon})/\sqrt{\alpha_t}$ into (6). The substitution turns the $\boldsymbol{x}_0$ term into a linear combination of $\boldsymbol{x}_t$ and $\boldsymbol{\epsilon}$, which then merges with the pre-existing $\boldsymbol{x}_t$ term in (6). Collecting coefficients

$$\tilde{\mu}_t(\boldsymbol{x}_t, \boldsymbol{x}_0(\boldsymbol{x}_t, \boldsymbol{\epsilon})) \;=\; \frac{1}{\sqrt{\alpha_t/\alpha_{t-1}}}\left(\boldsymbol{x}_t \;-\; \frac{\beta_t}{\sqrt{1-\alpha_t}}\,\boldsymbol{\epsilon}\right) \tag{8}$$

This suggests usage of noise-prediction parametrisation: instead of letting the network output an arbitrary mean, we fix

$$\boxed{\mu_\theta^{(t)}(\boldsymbol{x}_t) \;:=\; \frac{1}{\sqrt{\alpha_t/\alpha_{t-1}}}\left(\boldsymbol{x}_t \;-\; \frac{\beta_t}{\sqrt{1-\alpha_t}}\,\boldsymbol{\epsilon}_\theta^{(t)}(\boldsymbol{x}_t)\right)} \tag{9}$$

so the only learnable quantity is the noise estimate $\boldsymbol{\epsilon}_\theta^{(t)}(\boldsymbol{x}_t)$.

The KL term at step t in (5) compares two Gaussians with the same variance $\sigma_t^2 \boldsymbol{I}$, which reduces to a squared-error between their means

$$D_{\mathrm{KL}}\big(q(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0) \,\|\, p_\theta^{(t)}(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t)\big) \;=\; \frac{1}{2\sigma_t^2}\,\big\|\tilde{\mu}_t(\boldsymbol{x}_t, \boldsymbol{x}_0) - \mu_\theta^{(t)}(\boldsymbol{x}_t)\big\|_2^2 \;+\; \text{const} \tag{10}$$

Plugging (8) and (9) into the right-hand side, the $\boldsymbol{x}_t$ terms cancel and only the noise residual $\big\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta^{(t)}(\boldsymbol{x}_t)\big\|_2^2$ remains, scaled by a t-dependent positive weight. Summing over t and taking expectations, the ELBO collapses to a weighted sum of denoising losses

$$\boxed{L_\gamma(\theta) \;:=\; \sum_{t=1}^{T} \gamma_t\, \mathbb{E}_{\boldsymbol{x}_0 \sim q(\boldsymbol{x}_0),\, \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})}\Big[\big\|\boldsymbol{\epsilon}_\theta^{(t)}\big(\sqrt{\alpha_t}\,\boldsymbol{x}_0 + \sqrt{1-\alpha_t}\,\boldsymbol{\epsilon}\big) - \boldsymbol{\epsilon}\big\|_2^2\Big]} \tag{11}$$

where the positive weights γ_t are determined by the schedule $\alpha_{1:T}$ and the reverse variances $\sigma_{1:T}$. The special case $\gamma_t \equiv 1$ gives the unweighted loss

$$L_1(\theta) \;=\; \sum_{t=1}^{T} \mathbb{E}_{\boldsymbol{x}_0, \boldsymbol{\epsilon}}\big[\|\boldsymbol{\epsilon}_\theta^{(t)}(\boldsymbol{x}_t) - \boldsymbol{\epsilon}\|_2^2\big] \tag{12}$$

which Ho et al. [2020] found to give the best generation quality in practice. The observation on which the entire DDIM construction rests is that each term of L_γ depends only on the marginal $q(\boldsymbol{x}_t \mid \boldsymbol{x}_0)$ and not on the full joint $q(\boldsymbol{x}_{1:T} \mid \boldsymbol{x}_0)$.

3.2 A Non-Markovian Forward Process

The invariance of L_γ under marginal-preserving reparametrisations of the inference joint suggests we look for richer factorisations. In this section we construct a family of inference joints that share the marginals of Equation (2) but factorise differently, and verify the consistency of the construction.

For a vector hyperparameter $\boldsymbol{\sigma} \in \mathbb{R}_{\geq 0}^T$ we define the inference distribution

$$q_\sigma(\boldsymbol{x}_{1:T} \mid \boldsymbol{x}_0) \;=\; q_\sigma(\boldsymbol{x}_T \mid \boldsymbol{x}_0) \prod_{t=2}^{T} q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0) \tag{13}$$

with $q_\sigma(\boldsymbol{x}_T \mid \boldsymbol{x}_0) = \mathcal{N}(\sqrt{\alpha_T}\,\boldsymbol{x}_0, (1-\alpha_T)\boldsymbol{I})$ and, for $t > 1$,

$$q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0) \;=\; \mathcal{N}\!\left(\sqrt{\alpha_{t-1}}\,\boldsymbol{x}_0 \;+\; \sqrt{1-\alpha_{t-1}-\sigma_t^2}\;\cdot\;\frac{\boldsymbol{x}_t - \sqrt{\alpha_t}\,\boldsymbol{x}_0}{\sqrt{1-\alpha_t}},\; \sigma_t^2\boldsymbol{I}\right) \tag{14}$$

The form of the mean in (14) is chosen so that the marginals continue to match Equation (2). The next lemma makes this consistency property precise.

Lemma 1. *For q_σ defined in* (13)-(14)*, $q_\sigma(\boldsymbol{x}_t \mid \boldsymbol{x}_0) = \mathcal{N}(\sqrt{\alpha_t}\,\boldsymbol{x}_0,\, (1-\alpha_t)\boldsymbol{I})$ for all $t \in \{1, \ldots, T\}$.*

The proof is by backward induction in t, using the Gaussian linear-conditional identity [Bishop and Nasrabadi, 2006, Eq. 2.115] to integrate $\boldsymbol{x}_t$ out of the product after applying bayes rule. Full details in Appendix A.

Although (13)-(14) factorise the joint in reverse-time order, the "forward" conditional $q_\sigma(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}, \boldsymbol{x}_0)$ is recovered from Bayes' rule

$$q_\sigma(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}, \boldsymbol{x}_0) = \frac{q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0)\, q_\sigma(\boldsymbol{x}_t \mid \boldsymbol{x}_0)}{q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_0)} \tag{15}$$

which is again Gaussian since all three terms are.

Proposition 1 (Forward conditional)**.** *For each $t \geq 2$,*

$$q_\sigma(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}, \boldsymbol{x}_0) \;=\; \mathcal{N}\!\Big(\rho\,\boldsymbol{x}_{t-1} + \big(\sqrt{\alpha_t} - \rho\sqrt{\alpha_{t-1}}\big)\boldsymbol{x}_0,\;\; \tfrac{(1-\alpha_t)\sigma_t^2}{1-\alpha_{t-1}}\boldsymbol{I}\Big), \tag{16}$$

where $\rho := \sqrt{(1-\alpha_{t-1}-\sigma_t^2)(1-\alpha_t)}/(1-\alpha_{t-1})$.

The proof applies Bishop's reverse-conditional formula [Bishop and Nasrabadi, 2006, Eq. 2.116] to the linear-Gaussian system $(q_\sigma(\boldsymbol{x}_t \mid \boldsymbol{x}_0),\, q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0))$ from Lemma 1 and (14). Full details in Appendix A.

The mean of (16) carries an $\boldsymbol{x}_0$-coefficient $\sqrt{\alpha_t} - \rho\sqrt{\alpha_{t-1}}$, so in general the conditional depends on $\boldsymbol{x}_0$ as well as $\boldsymbol{x}_{t-1}$ and the inference process is genuinely non-Markovian. Song et al. [2020a] note that the DDPM chain (1) arises as the unique sub-family of q_σ in which this $\boldsymbol{x}_0$-coefficient vanishes. The opposite extreme $\sigma_t = 0$ yields the deterministic DDIM map, with all stochasticity removed from the chain conditional on $\boldsymbol{x}_T$.

Proposition 2 (DDPM-collapse condition)**.** *The conditional* (16) *is independent of $\boldsymbol{x}_0$ if and only if*

$$\sigma_t^2 \;=\; \tilde{\beta}_t \;:=\; \frac{1-\alpha_{t-1}}{1-\alpha_t}\,\big(1 - \alpha_t/\alpha_{t-1}\big) \tag{17}$$

With this choice, $q_\sigma(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}, \boldsymbol{x}_0) = \mathcal{N}\big(\sqrt{\alpha_t/\alpha_{t-1}}\,\boldsymbol{x}_{t-1},\, (1-\alpha_t/\alpha_{t-1})\boldsymbol{I}\big) = q(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1})$, recovering the DDPM forward kernel (1)*.*

The proof solves the iff condition $\rho\sqrt{\alpha_{t-1}} = \sqrt{\alpha_t}$ for σ_t^2 and substitutes back into (16). Full details in Appendix A.

3.3 The DDIM Generative Process

To initiate sampling, we draw $\boldsymbol{x}_T \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$. The reverse conditional $q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0)$ depends explicitly on the clean data $\boldsymbol{x}_0$, which is unavailable at sampling time. To turn it into a generative process we replace $\boldsymbol{x}_0$ by the network's best estimate. Since the noise prediction $\boldsymbol{\epsilon}_\theta^{(t)}(\boldsymbol{x}_t)$ approximates the noise $\boldsymbol{\epsilon}$ used to corrupt $\boldsymbol{x}_0$, inverting Equation (2) suggests the denoised estimate

$$f_\theta^{(t)}(\boldsymbol{x}_t) \;:=\; \frac{\boldsymbol{x}_t - \sqrt{1-\alpha_t}\,\boldsymbol{\epsilon}_\theta^{(t)}(\boldsymbol{x}_t)}{\sqrt{\alpha_t}} \qquad \text{(predicted } \boldsymbol{x}_0\text{)} \tag{18}$$

We then define the parametric reverse process by substituting $f_\theta^{(t)}(\boldsymbol{x}_t)$ for the unknown $\boldsymbol{x}_0$ in $q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0)$:

$$p_\theta^{(t)}(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t) = \begin{cases} \mathcal{N}\big(f_\theta^{(1)}(\boldsymbol{x}_1),\, \sigma_1^2\boldsymbol{I}\big) & t = 1 \\ q_\sigma\big(\boldsymbol{x}_{t-1} \,\big|\, \boldsymbol{x}_t,\, f_\theta^{(t)}(\boldsymbol{x}_t)\big) & t > 1 \end{cases} \tag{19}$$

The full generative model is $p_\theta(\boldsymbol{x}_{0:T}) = p_\theta(\boldsymbol{x}_T)\prod_{t=1}^{T} p_\theta^{(t)}(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t)$. Substituting $f_\theta^{(t)}(\boldsymbol{x}_t)$ into the mean of (14) and unfolding the Gaussian gives the explicit sampling update, valid for $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$ and the convention $\alpha_0 := 1$:

$$\boxed{\boldsymbol{x}_{t-1} \;=\; \sqrt{\alpha_{t-1}}\underbrace{\left(\frac{\boldsymbol{x}_t - \sqrt{1-\alpha_t}\,\boldsymbol{\epsilon}_\theta^{(t)}(\boldsymbol{x}_t)}{\sqrt{\alpha_t}}\right)}_{\text{“predicted } \boldsymbol{x}_0\text{”}} \;+\; \underbrace{\sqrt{1-\alpha_{t-1}-\sigma_t^2}\,\boldsymbol{\epsilon}_\theta^{(t)}(\boldsymbol{x}_t)}_{\text{“direction toward } \boldsymbol{x}_t\text{”}} \;+\; \underbrace{\sigma_t \boldsymbol{\epsilon}_t}_{\text{noise}}} \tag{20}$$

The three terms have a clean interpretation: the first is the rescaled denoised estimate, the second restores the deterministic component of the next noise level, and the third is the residual stochasticity controlled by σ_t.

The variance hyperparameter σ_t governs the stochasticity of the chain. Three choices are canonical, parameterised by a scalar $\eta \in [0, 1]$. Setting $\sigma_t = \sqrt{(1-\alpha_{t-1})/(1-\alpha_t)}\sqrt{1-\alpha_t/\alpha_{t-1}}$ ($\eta = 1$) recovers the DDPM Markovian reverse process. Setting $\sigma_t = 0$ for all t ($\eta = 0$) eliminates the noise term entirely, so $\boldsymbol{x}_{t-1}$ becomes a deterministic function of $\boldsymbol{x}_t$, giving us the DDIM update. Intermediate values $\sigma_t(\eta) = \eta\sqrt{(1-\alpha_{t-1})/(1-\alpha_t)}\sqrt{1-\alpha_t/\alpha_{t-1}}$ interpolate continuously between the two limits.

3.4 The Variational Objective

We now derive the variational objective associated with q_σ and show that, despite the new factorisation, it reduces to the familiar DDPM denoising loss up to a positive reweighting and a θ-independent constant. The analogue of the DDPM ELBO for the inference family q_σ with the parametric reverse p_θ defined in (19) is the negative variational bound

$$J_\sigma(\theta) \;=\; \mathbb{E}_{\boldsymbol{x}_{0:T}\sim q_\sigma}[\log q_\sigma(\boldsymbol{x}_{1:T} \mid \boldsymbol{x}_0) - \log p_\theta(\boldsymbol{x}_{0:T})] \tag{21}$$

Factorising q_σ as in (13) and p_θ with the chain rule, and dropping all terms independent of θ, gives the standard ELBO form

$$J_\sigma(\theta) \;\equiv\; \mathbb{E}_{\boldsymbol{x}_{0:T}\sim q_\sigma}\left[\sum_{t=2}^{T} D_{\mathrm{KL}}\big(q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0) \,\big\|\, p_\theta^{(t)}(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t)\big) \;-\; \log p_\theta^{(1)}(\boldsymbol{x}_0 \mid \boldsymbol{x}_1)\right] \tag{22}$$

where “$\equiv$” denotes equality up to an additive constant that depends on q_σ but not on θ. Each KL term contrasts the fixed reverse conditional $q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0)$, which uses the true $\boldsymbol{x}_0$, against the parametric reverse $p_\theta^{(t)}(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t)$, which uses the prediction $f_\theta^{(t)}(\boldsymbol{x}_t)$. The main theoretical result of Song et al. [2020a] is that this objective is the same as the DDPM objective L_γ for a particular choice of γ.

Theorem 1 (Song et al., 2020a, Theorem 1)**.** *For all $\boldsymbol{\sigma} \in \mathbb{R}_{>0}^T$ there exist weights $\gamma \in \mathbb{R}_{>0}^T$ and a constant $C \in \mathbb{R}$ such that*

$$J_\sigma(\theta) \;=\; L_\gamma(\theta) \;+\; C.$$

The proof computes each KL in (22) using the equal-variance Gaussian KL formula and exploits the cancellation of shift terms inherited from (14). Full details can be read in Appendix A. Song et al. [2020a] state $\gamma_t = 1/(2d\,\sigma_t^2\,\alpha_t)$, with d the data dimension, but our derivation yields $\gamma_t = c_t^2(1-\alpha_t)/(2\sigma_t^2\,\alpha_t)$ for $t \geq 2$, with $c_t := \sqrt{\alpha_{t-1}} - \sqrt{\alpha_t(1-\alpha_{t-1}-\sigma_t^2)/\sqrt{1-\alpha_t}}$. The discrepancies are of three kinds. The factor $(1-\alpha_t)$ comes inevitably from the noise-substitution $\boldsymbol{x}_0 - f_\theta^{(t)}(\boldsymbol{x}_t) = \frac{\sqrt{1-\alpha_t}}{\sqrt{\alpha_t}}(\boldsymbol{\epsilon}_\theta^{(t)} - \boldsymbol{\epsilon})$. The factor c_t^2 is the squared $\boldsymbol{x}_0$-coefficient of the affine mean of (14), which the equal-variance Gaussian KL leaves intact; it equals 1 only in the DDPM-collapse case, not generically. The factor $1/d$, on the other hand, can be reconciled by reading the paper's loss as a per-coordinate MSE rather than a sum-of-squares, although the paper does not make this convention explicit. Fortunately, none of these discrepancies affect the main qualitative claim of the theorem: without weight-sharing, every positive γ has the same minimiser as L_1, so the conclusion that the L_1-trained network is optimal across the $\boldsymbol{\sigma}$-family holds regardless of which closed form is correct. Full details in Appendix A.

The practical force of Theorem 1 comes from a simple optimisation observation. If the parameters $\theta = \{\theta^{(t)}\}_{t=1}^T$ are not shared across t, each summand $\gamma_t \mathbb{E}\|\boldsymbol{\epsilon}_\theta^{(t)}(\boldsymbol{x}_t) - \boldsymbol{\epsilon}\|_2^2$ is optimised independently

of the others, and multiplying it by a positive scalar leaves its minimiser unchanged. The minimiser of L_γ therefore coincides with the minimiser of L_1 for every choice of positive weights γ, and in particular for the weights produced by Theorem 1. A network trained with L_1 (a vanilla pretrained DDPM) is automatically optimal for the variational objective of every non-Markovian member of the family q_σ, and no retraining is required to obtain DDIM samples.

The theorem as stated requires $\boldsymbol{\sigma} > 0$, because each KL in (22) is undefined when one of the two Gaussians collapses to a delta. The deterministic case $\boldsymbol{\sigma} = 0$ is therefore not covered by the literal statement and must be understood as the limit $\sigma_t \to 0^+$. The sampling rule (20) extends continuously to this limit since this case can always be approximated by taking σ_t very small.

3.5 Accelerated Sampling

Theorem 1 already buys us deterministic sampling from a pretrained DDPM, but the sampling chain still has T steps. We now exploit the same marginal-only dependence of L_1 to shorten the chain itself. The idea is to define the inference process not over the full index set $\{1, \dots, T\}$ but over a much shorter sub-sequence $\tau = (\tau_1, \dots, \tau_S)$, with $S \ll T$ and $\tau_S = T$. Letting $\bar{\tau} = \{1, \dots, T\} \setminus \tau$ denote the complement, we define the joint

$$q_{\sigma,\tau}(\boldsymbol{x}_{1:T} \mid \boldsymbol{x}_0) \;=\; q_{\sigma,\tau}(\boldsymbol{x}_{\tau_S} \mid \boldsymbol{x}_0) \prod_{i=1}^{S} q_{\sigma,\tau}(\boldsymbol{x}_{\tau_{i-1}} \mid \boldsymbol{x}_{\tau_i}, \boldsymbol{x}_0) \prod_{t \in \bar{\tau}} q_{\sigma,\tau}(\boldsymbol{x}_t \mid \boldsymbol{x}_0) \tag{23}$$

with $q_{\sigma,\tau}(\boldsymbol{x}_t \mid \boldsymbol{x}_0) = \mathcal{N}(\sqrt{\alpha_t}\boldsymbol{x}_0,\, (1-\alpha_t)\boldsymbol{I})$ for $t \in \bar{\tau} \cup \{T\}$, and the chain conditionals defined as in (14) but on the index set τ:

$$q_{\sigma,\tau}(\boldsymbol{x}_{\tau_{i-1}} \mid \boldsymbol{x}_{\tau_i}, \boldsymbol{x}_0) = \mathcal{N}\!\left(\sqrt{\alpha_{\tau_{i-1}}}\boldsymbol{x}_0 + \sqrt{1-\alpha_{\tau_{i-1}}-\sigma_{\tau_i}^2}\,\frac{\boldsymbol{x}_{\tau_i} - \sqrt{\alpha_{\tau_i}}\boldsymbol{x}_0}{\sqrt{1-\alpha_{\tau_i}}},\; \sigma_{\tau_i}^2\boldsymbol{I}\right) \tag{24}$$

A short calculation analogous to Lemma 1 shows that the marginals are still given by $q_{\sigma,\tau}(\boldsymbol{x}_t \mid \boldsymbol{x}_0) = \mathcal{N}(\sqrt{\alpha_t}\boldsymbol{x}_0, (1-\alpha_t)\boldsymbol{I})$ for every t.

We then define a generative process that uses only the chain indices to produce samples and assigns the remaining indices to a denoising-as-prediction term that appears only in the objective

$$p_\theta(\boldsymbol{x}_{0:T}) \;=\; p_\theta(\boldsymbol{x}_T) \underbrace{\prod_{i=1}^{S} p_\theta^{(\tau_i)}(\boldsymbol{x}_{\tau_{i-1}} \mid \boldsymbol{x}_{\tau_i})}_{\text{used to produce samples}} \underbrace{\prod_{t \in \bar{\tau}} p_\theta^{(t)}(\boldsymbol{x}_0 \mid \boldsymbol{x}_t)}_{\text{only in the objective}} \tag{25}$$

where for the chain factors $p_\theta^{(\tau_i)}(\boldsymbol{x}_{\tau_{i-1}} \mid \boldsymbol{x}_{\tau_i}) = q_{\sigma,\tau}(\boldsymbol{x}_{\tau_{i-1}} \mid \boldsymbol{x}_{\tau_i},\, f_\theta^{(\tau_i)}(\boldsymbol{x}_{\tau_i}))$ for $i > 1$, and for the star factors $p_\theta^{(t)}(\boldsymbol{x}_0 \mid \boldsymbol{x}_t) = \mathcal{N}(f_\theta^{(t)}(\boldsymbol{x}_t),\, \sigma_t^2\boldsymbol{I})$. The associated variational objective decomposes into a chain part and a star part

$$\begin{aligned} J_{\sigma,\tau}(\theta) \;\equiv\; \mathbb{E}_{\boldsymbol{x}_{0:T} \sim q_{\sigma,\tau}}\Bigg[& \sum_{t \in \bar{\tau}} D_{\mathrm{KL}}\big(q_{\sigma,\tau}(\boldsymbol{x}_t \mid \boldsymbol{x}_0) \,\big\|\, p_\theta^{(t)}(\boldsymbol{x}_0 \mid \boldsymbol{x}_t)\big) \\ & + \sum_{i=1}^{S} D_{\mathrm{KL}}\big(q_{\sigma,\tau}(\boldsymbol{x}_{\tau_{i-1}} \mid \boldsymbol{x}_{\tau_i}, \boldsymbol{x}_0) \,\big\|\, p_\theta^{(\tau_i)}(\boldsymbol{x}_{\tau_{i-1}} \mid \boldsymbol{x}_{\tau_i})\big)\Bigg] \end{aligned} \tag{26}$$

Each KL term is between two Gaussians with θ-independent variances, and repeating the calculation of Theorem 1 on each term shows that $J_{\sigma,\tau}(\theta) = L_{\gamma'}(\theta) + C'$ for some positive weights $\gamma' \in \mathbb{R}^T_{>0}$. The chain terms supply weights at the indices in τ and the star terms at the indices in $\bar{\tau}$, so together they cover every t. By the same independence argument as before, the minimiser is still the L_1 optimum, and the same pretrained $\boldsymbol{\epsilon}_\theta$ may be reused for the accelerated process without retraining.

Restricting Equation (20) to the chain indices and using the convention $\alpha_{\tau_0} := 1$ gives the explicit accelerated update

$$\boxed{\boldsymbol{x}_{\tau_{i-1}} \;=\; \sqrt{\alpha_{\tau_{i-1}}}\left(\frac{\boldsymbol{x}_{\tau_i} - \sqrt{1-\alpha_{\tau_i}}\,\boldsymbol{\epsilon}_\theta^{(\tau_i)}(\boldsymbol{x}_{\tau_i})}{\sqrt{\alpha_{\tau_i}}}\right) \;+\; \sqrt{1-\alpha_{\tau_{i-1}}-\sigma_{\tau_i}^2(\eta)}\,\boldsymbol{\epsilon}_\theta^{(\tau_i)}(\boldsymbol{x}_{\tau_i}) \;+\; \sigma_{\tau_i}(\eta)\,\boldsymbol{\epsilon}} \tag{27}$$

The total sampling cost is S network evaluations. Song et al. [2020a] report results with two simple sub-sequence schemes: a linear schedule $\tau_i = \lfloor c\, i \rfloor$ used on LSUN and CelebA, and a quadratic schedule $\tau_i = \lfloor c\, i^2 \rfloor$ used on CIFAR-10, with the constant c chosen so that $\tau_S = T$.

4 Experiments and Discussion

4.1 Setup

Song et al. [2020a] keep the training side identical to Ho et al. [2020] and vary only the sampling procedure. The same noise-prediction network $\boldsymbol{\epsilon}_\theta$ is used throughout, trained with the unweighted L_1 loss (12) for $T = 1000$ steps under the linear β-schedule of Ho et al. [2020]. Sampling is done over a sub-sequence τ of length $S \in \{10, 20, 50, 100, 1000\}$, with two simple schedules: linear $\tau_i = \lfloor c\, i \rfloor$ on CelebA and LSUN, and quadratic $\tau_i = \lfloor c\, i^2 \rfloor$ on CIFAR-10. The stochasticity is parametrised by $\eta \in \{0, 0.2, 0.5, 1.0\}$ together with the high-noise DDPM variant $\hat{\sigma}_{\tau_i} = \sqrt{1 - \alpha_{\tau_i} / \alpha_{\tau_{i-1}}}$. All FID numbers reported here come directly from Song et al. [2020a].

4.2 Sample quality

Table 1 reproduces the FID figures from Song et al. [2020a, Table 1] on CIFAR-10 (32×32) and CelebA (64×64) with the same DDPM-trained network.

Table 1: CIFAR10 and CelebA image generation measured in FID. $\eta = 1.0$ and $\hat{\sigma}$ are cases of DDPM (although Ho et al. (2020) only considered $T = 1000$ steps, and $S < T$ can be seen as simulating DDPMs trained with S steps), and $\eta = 0.0$ indicates DDIM.

		CIFAR10 (32 × 32)					CelebA (64 × 64)				
	S	10	20	50	100	1000	10	20	50	100	1000
η	0.0	**13.36**	**6.84**	**4.67**	**4.16**	4.04	**17.33**	**13.73**	**9.17**	**6.53**	3.51
	0.2	14.04	7.11	4.77	4.25	4.09	17.66	14.11	9.51	6.79	3.64
	0.5	16.66	8.35	5.25	4.46	4.29	19.86	16.06	11.01	8.09	4.28
	1.0	41.07	18.36	8.01	5.78	4.73	33.12	26.03	18.48	13.93	5.98
$\hat{\sigma}$		367.43	133.37	32.72	9.99	**3.17**	299.71	183.83	71.71	45.20	**3.26**

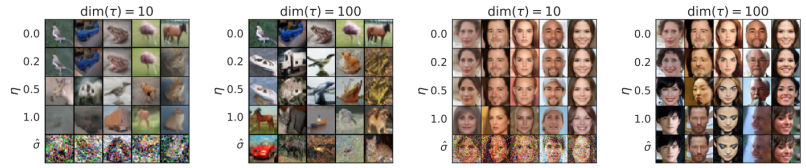


Figure 3: CIFAR10 and CelebA samples with $\dim(\tau) = 10$ and $\dim(\tau) = 100$.

Figure 1: FID on CIFAR-10 and CelebA. Rows: stochasticity $\eta \in \{0, 0.2, 0.5, 1.0\}$ and $\hat{\sigma}$ (the $\hat{\sigma}_{\tau_i} = \sqrt{1 - \alpha_{\tau_i} / \alpha_{\tau_{i-1}}}$ DDPM variant of Ho et al., 2020). Columns: number of sampling steps S.

DDIM ($\eta = 0$) dominates at $S \leq 100$ steps, while the high-noise variant $\hat{\sigma}$ is best at $S = 1000$ but degrades sharply for small S. On CelebA, the FID of 100-step DDPM ($\eta = 1$) is roughly matched by 20-step DDIM, a $5\times$ speedup at comparable quality. A similar pattern can be seen in CIFAR-10. The $\hat{\sigma}$ row also illustrates that the original DDPM variance choice is fundamentally tuned for the full $T = 1000$ chain.

4.3 Sample consistency and reconstruction

Because DDIM's reverse process is deterministic, the output $\boldsymbol{x}_0$ is a function of $\boldsymbol{x}_T$ alone. Song et al. [2020a] observe that fixing $\boldsymbol{x}_T$ and varying S yields images with the same high-level features (pose, identity, layout) but increasingly fine detail at larger S. This is shown in Figure 2: $\boldsymbol{x}_T$ encodes semantics, while the trajectory length S controls detail. DDPM cannot do this; different runs from the same $\boldsymbol{x}_T$ produce visibly different images.

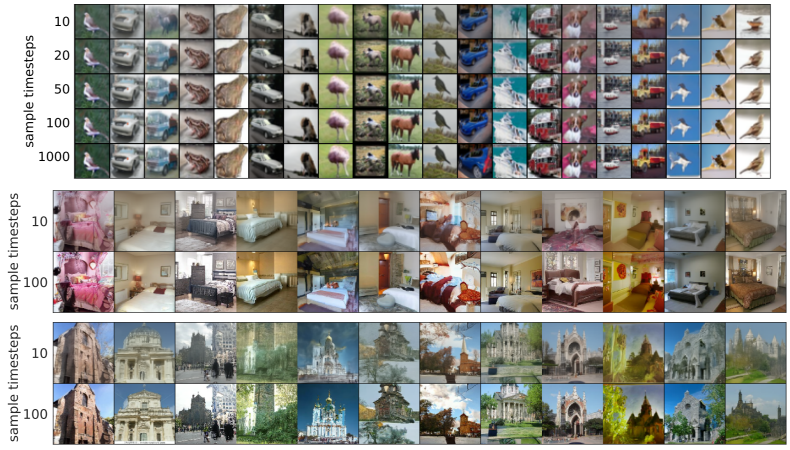


Figure 5: Samples from DDIM with the same random $\boldsymbol{x}_T$ and different number of steps.

Figure 2: Sample consistency: same $\boldsymbol{x}_T$, varying S [Song et al., 2020a]. High-level features remain stable across trajectory lengths.

4.4 Latent-space interpolation

Because $\boldsymbol{x}_T$ is a meaningful latent code, one can interpolate between two latents $\boldsymbol{x}_T^{(0)}, \boldsymbol{x}_T^{(1)}$ using spherical linear interpolation [Shoemake, 1985] $\boldsymbol{x}_T^{(\lambda)} = \frac{\sin((1-\lambda)\phi)}{\sin\phi}\boldsymbol{x}_T^{(0)} + \frac{\sin(\lambda\phi)}{\sin\phi}\boldsymbol{x}_T^{(1)}$ with $\phi = \arccos\big((\boldsymbol{x}_T^{(0)})^\top \boldsymbol{x}_T^{(1)} / (\|\boldsymbol{x}_T^{(0)}\|\|\boldsymbol{x}_T^{(1)}\|)\big)$. The slerp parametrisation is used in place of linear interpolation because $\boldsymbol{x}_T$ is approximately Gaussian and concentrated on a sphere of radius $\sqrt{d}$, so straight-line interpolants pass through low-density regions and produce blurry intermediates. Spherical interpolation stays on the sphere and yields semantically smooth transitions on CelebA and LSUN Church [Figs. 6-7; Song et al., 2020a]. DDPMs cannot interpolate this way because their stochastic reverse decouples $\boldsymbol{x}_T$ from $\boldsymbol{x}_0$.

5 Limitations

The framework has several limitations that are worth flagging. First, Theorem 1 is stated for $\boldsymbol{\sigma} > 0$, since the proof relies on the KL between two Gaussians of strictly positive variance. The deterministic case $\boldsymbol{\sigma} = 0$, which is the DDIM proper, is therefore reached only as a limit; a separate justification through the probability-flow ODE (which I havent completely understood) is needed to argue that the L_1-trained network remains optimal at the boundary.

Second, the practical claim that a vanilla pretrained DDPM is optimal for every member of the family q_σ relies on the not-shared-across-t assumption used in the proof (each γ_t summand is optimised independently). Standard DDPM architectures share a single U-Net across all t via a positional embedding, so the per-t minimisers are coupled in practice. The empirical observation that this coupling does not degrade DDIM samples is unproven, and we are not aware of a theoretical guarantee.

Third, sample quality from a pretrained DDPM still degrades for very small S. Although DDIM matches 100-step DDPM with roughly 20 steps on CelebA, FID worsens noticeably at $S \leq 10$.

Fourth, the construction is tied to a Gaussian forward process. Discrete data and other non-Gaussian targets require separate constructions and are outside the scope of the present derivation.

Finally, the choice of sub-sequence τ in the accelerated chain is a hyperparameter. Song et al. [2020a] report results for linear and quadratic schedules, but no closed-form optimal τ is known.

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A Detailed Proofs

Proof of Lemma 1. Backward induction on $t \in \{1, \dots, T\}$.

Base case ($t = T$). By the boundary condition in (13), $q_\sigma(\boldsymbol{x}_T \mid \boldsymbol{x}_0) = \mathcal{N}(\sqrt{\alpha_T}\boldsymbol{x}_0, (1-\alpha_T)\boldsymbol{I})$, which matches the claim.

Inductive step. Fix $t \geq 2$ and assume $q_\sigma(\boldsymbol{x}_t \mid \boldsymbol{x}_0) = \mathcal{N}(\sqrt{\alpha_t}\boldsymbol{x}_0, (1-\alpha_t)\boldsymbol{I})$. By the factorisation (13),

$$q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_0) \;=\; \int q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0)\, q_\sigma(\boldsymbol{x}_t \mid \boldsymbol{x}_0)\, d\boldsymbol{x}_t.$$

We use the linear-Gaussian marginalisation identity [Bishop and Nasrabadi, 2006, Eq. 2.115]: if $p(\boldsymbol{x}) = \mathcal{N}(\boldsymbol{x};\, \mu_a, \Sigma_a)$ and $p(\boldsymbol{y} \mid \boldsymbol{x}) = \mathcal{N}(\boldsymbol{y};\, A\boldsymbol{x} + \boldsymbol{b},\, L)$, then

$$\int p(\boldsymbol{y} \mid \boldsymbol{x})\, p(\boldsymbol{x})\, d\boldsymbol{x} \;=\; \mathcal{N}\big(\boldsymbol{y};\, A\mu_a + \boldsymbol{b},\; L + A\Sigma_a A^\top\big).$$

Identifying $\boldsymbol{x} \leftrightarrow \boldsymbol{x}_t$, $\boldsymbol{y} \leftrightarrow \boldsymbol{x}_{t-1}$, and reading off the parameters of (14),

$$A = \tfrac{\sqrt{1-\alpha_{t-1}-\sigma_t^2}}{\sqrt{1-\alpha_t}}\boldsymbol{I}, \quad \boldsymbol{b} = \sqrt{\alpha_{t-1}}\boldsymbol{x}_0 - A\sqrt{\alpha_t}\boldsymbol{x}_0, \quad L = \sigma_t^2\boldsymbol{I}, \quad \mu_a = \sqrt{\alpha_t}\boldsymbol{x}_0, \quad \Sigma_a = (1-\alpha_t)\boldsymbol{I}.$$

The output mean is

$$A\mu_a + \boldsymbol{b} \;=\; A\sqrt{\alpha_t}\boldsymbol{x}_0 + \sqrt{\alpha_{t-1}}\boldsymbol{x}_0 - A\sqrt{\alpha_t}\boldsymbol{x}_0 \;=\; \sqrt{\alpha_{t-1}}\boldsymbol{x}_0,$$

and the output covariance is

$$L + A\Sigma_a A^\top \;=\; \sigma_t^2\boldsymbol{I} + \tfrac{1-\alpha_{t-1}-\sigma_t^2}{1-\alpha_t}(1-\alpha_t)\boldsymbol{I} \;=\; \sigma_t^2\boldsymbol{I} + (1-\alpha_{t-1}-\sigma_t^2)\boldsymbol{I} \;=\; (1-\alpha_{t-1})\boldsymbol{I}.$$

Hence $q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_0) = \mathcal{N}(\sqrt{\alpha_{t-1}}\boldsymbol{x}_0,\, (1-\alpha_{t-1})\boldsymbol{I})$, completing the induction. □

Proof of Proposition 1. Fix $t \geq 2$. Use the same identification as in the proof of Lemma 1: $p(\boldsymbol{x}) := q_\sigma(\boldsymbol{x}_t \mid \boldsymbol{x}_0) = \mathcal{N}(\boldsymbol{x};\, \mu_a, \Sigma_a)$ and $p(\boldsymbol{y} \mid \boldsymbol{x}) := q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0) = \mathcal{N}(\boldsymbol{y};\, A\boldsymbol{x} + \boldsymbol{b},\, L)$, with

$$\mu_a = \sqrt{\alpha_t}\boldsymbol{x}_0, \qquad \Sigma_a = (1-\alpha_t)\boldsymbol{I}, \qquad A = \tfrac{\sqrt{1-\alpha_{t-1}-\sigma_t^2}}{\sqrt{1-\alpha_t}}\boldsymbol{I} =: a\boldsymbol{I}$$

$$\boldsymbol{b} = \sqrt{\alpha_{t-1}}\boldsymbol{x}_0 - A\sqrt{\alpha_t}\boldsymbol{x}_0 = (\sqrt{\alpha_{t-1}} - a\sqrt{\alpha_t})\boldsymbol{x}_0, \qquad L = \sigma_t^2\boldsymbol{I}.$$

What we want is the Bayes-flip $p(\boldsymbol{x} \mid \boldsymbol{y}) = q_\sigma(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}, \boldsymbol{x}_0)$. Bishop's reverse-conditional formula [Bishop and Nasrabadi, 2006, Eq. 2.116], in the same notation, gives $p(\boldsymbol{x} \mid \boldsymbol{y}) = \mathcal{N}(\boldsymbol{x};\, \boldsymbol{m},\, \Sigma)$ with

$$\boldsymbol{m} \;:=\; \Sigma\big[A^\top L^{-1}(\boldsymbol{y}-\boldsymbol{b}) + \Sigma_a^{-1}\mu_a\big], \qquad \Sigma \;:=\; \big(\Sigma_a^{-1} + A^\top L^{-1} A\big)^{-1}.$$

Posterior covariance. With $A = a\boldsymbol{I}$, $L = \sigma_t^2 \boldsymbol{I}$, $\Sigma_a = (1-\alpha_t)\boldsymbol{I}$,

$$\Sigma_a^{-1} + A^\top L^{-1} A \;=\; \left(\tfrac{1}{1-\alpha_t} + \tfrac{a^2}{\sigma_t^2}\right)\boldsymbol{I} \;=\; \tfrac{\sigma_t^2 + a^2(1-\alpha_t)}{(1-\alpha_t)\sigma_t^2}\boldsymbol{I}$$

Using $a^2(1-\alpha_t) = 1 - \alpha_{t-1} - \sigma_t^2$, the numerator simplifies to $1-\alpha_{t-1}$, hence

$$\Sigma \;=\; \tfrac{(1-\alpha_t)\sigma_t^2}{1-\alpha_{t-1}}\boldsymbol{I}$$

Posterior mean. $A^\top L^{-1}(\boldsymbol{y}-\boldsymbol{b}) = (a/\sigma_t^2)(\boldsymbol{x}_{t-1} - \boldsymbol{b})$ and $\Sigma_a^{-1}\mu_a = \sqrt{\alpha_t}/(1-\alpha_t)\,\boldsymbol{x}_0$, so

$$\boldsymbol{m} \;=\; \Sigma\big[A^\top L^{-1}(\boldsymbol{y}-\boldsymbol{b}) + \Sigma_a^{-1}\mu_a\big] \;=\; \underbrace{\tfrac{a(1-\alpha_t)}{1-\alpha_{t-1}}}_{=:\,\rho}(\boldsymbol{x}_{t-1} - \boldsymbol{b}) \;+\; \tfrac{\sigma_t^2\sqrt{\alpha_t}}{1-\alpha_{t-1}}\boldsymbol{x}_0,$$

with $\rho = \sqrt{(1-\alpha_{t-1}-\sigma_t^2)(1-\alpha_t)}/(1-\alpha_{t-1})$. Substituting $\boldsymbol{b}$ and collecting by variable,

$$\begin{aligned}\boldsymbol{m} \;&=\; \rho\,\boldsymbol{x}_{t-1} \;-\; \rho\sqrt{\alpha_{t-1}}\,\boldsymbol{x}_0 \;+\; \rho a\sqrt{\alpha_t}\,\boldsymbol{x}_0 \;+\; \tfrac{\sigma_t^2\sqrt{\alpha_t}}{1-\alpha_{t-1}}\boldsymbol{x}_0 \\ &=\; \rho\,\boldsymbol{x}_{t-1} \;-\; \rho\sqrt{\alpha_{t-1}}\,\boldsymbol{x}_0 \;+\; \left(\rho a + \tfrac{\sigma_t^2}{1-\alpha_{t-1}}\right)\sqrt{\alpha_t}\,\boldsymbol{x}_0\end{aligned}$$

Now $\rho a = a^2(1-\alpha_t)/(1-\alpha_{t-1}) = (1-\alpha_{t-1}-\sigma_t^2)/(1-\alpha_{t-1})$, so $\rho a + \sigma_t^2/(1-\alpha_{t-1}) = 1$, giving

$$\boldsymbol{m} \;=\; \rho\,\boldsymbol{x}_{t-1} \;+\; \big(\sqrt{\alpha_t} - \rho\sqrt{\alpha_{t-1}}\big)\boldsymbol{x}_0,$$

which is exactly the mean stated in (16). □

Proof of Proposition 2. By Proposition 1, the mean of $q_\sigma(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}, \boldsymbol{x}_0)$ has $\boldsymbol{x}_0$-coefficient $\sqrt{\alpha_t} - \rho\sqrt{\alpha_{t-1}}$. It vanishes iff $\rho\sqrt{\alpha_{t-1}} = \sqrt{\alpha_t}$, iff $\rho^2 = \alpha_t/\alpha_{t-1}$. Squaring the definition of ρ and solving for σ_t^2:

$$\tfrac{(1-\alpha_{t-1}-\sigma_t^2)(1-\alpha_t)}{(1-\alpha_{t-1})^2} = \tfrac{\alpha_t}{\alpha_{t-1}} \iff \sigma_t^2 \;=\; (1-\alpha_{t-1}) - \tfrac{\alpha_t(1-\alpha_{t-1})^2}{\alpha_{t-1}(1-\alpha_t)}.$$

Factoring,

$$\sigma_t^2 \;=\; \tfrac{1-\alpha_{t-1}}{\alpha_{t-1}(1-\alpha_t)}\big[\alpha_{t-1}(1-\alpha_t) - \alpha_t(1-\alpha_{t-1})\big] \;=\; \tfrac{(1-\alpha_{t-1})(\alpha_{t-1}-\alpha_t)}{\alpha_{t-1}(1-\alpha_t)} \;=\; \tfrac{1-\alpha_{t-1}}{1-\alpha_t}\big(1-\alpha_t/\alpha_{t-1}\big) \;=\; \tilde{\beta}_t.$$

With $\sigma_t^2 = \tilde{\beta}_t$, $\rho = \sqrt{\alpha_t/\alpha_{t-1}}$, so the mean of (16) reduces to $\sqrt{\alpha_t/\alpha_{t-1}}\,\boldsymbol{x}_{t-1}$, and the covariance becomes

$$\tfrac{(1-\alpha_t)\tilde{\beta}_t}{1-\alpha_{t-1}}\boldsymbol{I} \;=\; \tfrac{(1-\alpha_t)}{1-\alpha_{t-1}} \cdot \tfrac{1-\alpha_{t-1}}{1-\alpha_t}\big(1-\alpha_t/\alpha_{t-1}\big)\boldsymbol{I} \;=\; \big(1-\alpha_t/\alpha_{t-1}\big)\boldsymbol{I},$$

which is precisely the DDPM forward kernel $q(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1})$ of (1). □

Proof of Theorem 1. Substituting the factorisation (13) of q_σ and the chain-rule factorisation of p_θ into the definition (21),

$$\begin{aligned}J_\sigma(\theta) = \mathbb{E}_{q_\sigma}\big[\log q_\sigma(\boldsymbol{x}_T \mid \boldsymbol{x}_0) - \log p_\theta(\boldsymbol{x}_T)\big] \\ + \sum_{t=2}^{T} \mathbb{E}_{q_\sigma}\big[\log q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0) - \log p_\theta^{(t)}(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t)\big] \;-\; \mathbb{E}_{q_\sigma}\big[\log p_\theta^{(1)}(\boldsymbol{x}_0 \mid \boldsymbol{x}_1)\big].\end{aligned}$$

The first line equals $\mathbb{E}_{q(\boldsymbol{x}_0)} D_{\mathrm{KL}}\big(q_\sigma(\boldsymbol{x}_T \mid \boldsymbol{x}_0) \,\|\, p_\theta(\boldsymbol{x}_T)\big)$, which is constant in θ (both arguments are fixed Gaussians) and is absorbed into the constant C. For each $t \geq 2$, the term inside the summation depends only on $(\boldsymbol{x}_0, \boldsymbol{x}_{t-1}, \boldsymbol{x}_t)$

$$J_\sigma(\theta) \;=\; \sum_{t=2}^{T} \mathbb{E}_{q_\sigma(\boldsymbol{x}_{t-1}, \boldsymbol{x}_t, \boldsymbol{x}_0)}\big[\log q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0) - \log p_\theta^{(t)}(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t)\big] \;-\; \mathbb{E}_{q_\sigma(\boldsymbol{x}_1, \boldsymbol{x}_0)}\big[\log p_\theta^{(1)}(\boldsymbol{x}_0 \mid \boldsymbol{x}_1)\big] \;+\; \text{const}$$

Factor the joint as $q_\sigma(\boldsymbol{x}_{t-1}, \boldsymbol{x}_t, \boldsymbol{x}_0) = q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0)\, q_\sigma(\boldsymbol{x}_t, \boldsymbol{x}_0)$. The integrand depends on $\boldsymbol{x}_{t-1}$ only through the two log-densities, so doing the $\boldsymbol{x}_{t-1}$-integral first against $q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0)$ produces a KL by definition:

$$\mathbb{E}_{q_\sigma(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t,\boldsymbol{x}_0)}\big[\log q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0) - \log p_\theta^{(t)}(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t)\big] \;=\; D_{\mathrm{KL}}\big(q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0) \,\|\, p_\theta^{(t)}(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t)\big)$$

Pushing this inside the sum and absorbing the remaining $\mathbb{E}_{q_\sigma(\boldsymbol{x}_t,\boldsymbol{x}_0)}$ outside,

$$J_\sigma(\theta) \;=\; \sum_{t=2}^{T} \mathbb{E}_{q_\sigma(\boldsymbol{x}_t,\boldsymbol{x}_0)}\Big[D_{\mathrm{KL}}\big(q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0) \,\|\, p_\theta^{(t)}(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t)\big)\Big] - \mathbb{E}_{q_\sigma(\boldsymbol{x}_1,\boldsymbol{x}_0)}\big[\log p_\theta^{(1)}(\boldsymbol{x}_0 \mid \boldsymbol{x}_1)\big] + \text{const}$$

Both $q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0)$ and $p_\theta^{(t)}(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t) = q_\sigma\big(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t,\, f_\theta^{(t)}(\boldsymbol{x}_t)\big)$ are Gaussians with the same variance $\sigma_t^2 \boldsymbol{I}$; their means are obtained by substituting $\boldsymbol{x}_0$ versus $\hat{\boldsymbol{x}}_0 := f_\theta^{(t)}(\boldsymbol{x}_t)$ into the affine map of (14). With $A_t := \sqrt{1-\alpha_{t-1}-\sigma_t^2}/\sqrt{1-\alpha_t}$ the affine map has the form $\mu(\boldsymbol{x}_t, \boldsymbol{x}_0) = (\sqrt{\alpha_{t-1}} - A_t\sqrt{\alpha_t})\boldsymbol{x}_0 + A_t \boldsymbol{x}_t$, so the $\boldsymbol{x}_t$-shift cancels under subtraction:

$$\mu(\boldsymbol{x}_t, \boldsymbol{x}_0) - \mu(\boldsymbol{x}_t, \hat{\boldsymbol{x}}_0) \;=\; c_t\,(\boldsymbol{x}_0 - \hat{\boldsymbol{x}}_0), \qquad c_t \;:=\; \sqrt{\alpha_{t-1}} - \frac{\sqrt{\alpha_t(1-\alpha_{t-1}-\sigma_t^2)}}{\sqrt{1-\alpha_t}}$$

The KL between equal-variance Gaussians is the scaled squared mean difference, so

$$D_{\mathrm{KL}}\big(q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0) \,\big\|\, p_\theta^{(t)}(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t)\big) \;=\; \tfrac{c_t^2}{2\sigma_t^2}\big\|\boldsymbol{x}_0 - f_\theta^{(t)}(\boldsymbol{x}_t)\big\|_2^2.$$

Substituting $\boldsymbol{x}_t = \sqrt{\alpha_t}\boldsymbol{x}_0 + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}$ from (2) into the definition (18) of $f_\theta^{(t)}$,

$$f_\theta^{(t)}(\boldsymbol{x}_t) \;=\; \tfrac{\boldsymbol{x}_t - \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_\theta^{(t)}(\boldsymbol{x}_t)}{\sqrt{\alpha_t}} \;=\; \boldsymbol{x}_0 \;+\; \tfrac{\sqrt{1-\alpha_t}}{\sqrt{\alpha_t}}\big(\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta^{(t)}(\boldsymbol{x}_t)\big)$$

Hence

$$\boldsymbol{x}_0 - f_\theta^{(t)}(\boldsymbol{x}_t) \;=\; \tfrac{\sqrt{1-\alpha_t}}{\sqrt{\alpha_t}}\big(\boldsymbol{\epsilon}_\theta^{(t)}(\boldsymbol{x}_t) - \boldsymbol{\epsilon}\big),$$

and substituting back above,

$$D_{\mathrm{KL}}\big(q_\sigma(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t, \boldsymbol{x}_0) \,\big\|\, p_\theta^{(t)}(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t)\big) \;=\; \tfrac{c_t^2(1-\alpha_t)}{2\sigma_t^2\,\alpha_t}\,\big\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta^{(t)}(\boldsymbol{x}_t)\big\|_2^2$$

The expectation over $q_\sigma(\boldsymbol{x}_t, \boldsymbol{x}_0)$ becomes one over $\boldsymbol{x}_0 \sim q(\boldsymbol{x}_0)$ and $\boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$.

For the last term, by definition (19), $p_\theta^{(1)}(\boldsymbol{x}_0 \mid \boldsymbol{x}_1) = \mathcal{N}\big(f_\theta^{(1)}(\boldsymbol{x}_1),\, \sigma_1^2 \boldsymbol{I}\big)$, so

$$-\log p_\theta^{(1)}(\boldsymbol{x}_0 \mid \boldsymbol{x}_1) \;=\; \tfrac{d}{2}\log(2\pi\sigma_1^2) \;+\; \tfrac{1}{2\sigma_1^2}\big\|\boldsymbol{x}_0 - f_\theta^{(1)}(\boldsymbol{x}_1)\big\|_2^2$$

The first term is constant in θ and absorbed into C. The second is handled exactly as before (with $t = 1$):

$$\tfrac{1}{2\sigma_1^2}\big\|\boldsymbol{x}_0 - f_\theta^{(1)}(\boldsymbol{x}_1)\big\|_2^2 \;=\; \tfrac{1-\alpha_1}{2\sigma_1^2\,\alpha_1}\,\big\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta^{(1)}(\boldsymbol{x}_1)\big\|_2^2.$$

Conclusion. Define

$$\gamma_1 \;:=\; \tfrac{1-\alpha_1}{2\sigma_1^2\,\alpha_1}, \qquad \gamma_t \;:=\; \tfrac{c_t^2(1-\alpha_t)}{2\sigma_t^2\,\alpha_t} \quad (t \geq 2).$$

Hence $\gamma_t > 0$ for every t, and

$$J_\sigma(\theta) \;=\; \sum_{t=1}^{T} \gamma_t\, \mathbb{E}_{\boldsymbol{x}_0 \sim q(\boldsymbol{x}_0),\, \boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})}\big[\big\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta^{(t)}(\boldsymbol{x}_t)\big\|_2^2\big] \;+\; C \;=\; L_\gamma(\theta) + C,$$

with $\boldsymbol{x}_t = \sqrt{\alpha_t}\boldsymbol{x}_0 + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}$ and $C \in \mathbb{R}$ collecting the dropped constants. □

B Answers to Unanswered Questions Raised in the Presentation

Q1. **How does the motivation align with making the process non-Markovian?** *(Addressed in Section 1 and Section 3.2.)*
The motivation is to skip steps at sample time. The Markovian assumption forces the joint inference distribution to be uniquely determined by single-step transitions, which in turn forces the reverse p_θ to take T steps. Going non-Markovian widens the family of admissible joints which gives us the freedom to choose a shorter chain that does not match the forward chain length.

Q2. **Are the NNs shared between the timesteps or not?** *(Addressed in Section 3.4 and Section 5.)*
In practice, shared. The standard DDPM architecture [Ho et al., 2020] is a single U-Net $\boldsymbol{\epsilon}_\theta(\boldsymbol{x}, t)$ in which t enters as a positional embedding, so all T time slots use the same parameters; the per-t notation $\boldsymbol{\epsilon}_\theta^{(t)}$ is bookkeeping. The proof of Theorem 1 however treats $\theta = \{\theta^{(t)}\}$ as T disjoint blocks, since that decoupling is what makes each γ_t summand insensitive to its scalar weight and gives L_γ and L_1 the same minimiser. Once weights are tied across t, the per-step minimisers are coupled and the argument no longer applies as written. Song et al. [2020a] do not patch this gap and rely on the empirical observation that the same L_1-trained network produces good samples across the entire $\boldsymbol{\sigma}$-family.

Q3. **What happens with the last (star) term in the factorised forward process for accelerated sampling?** *(Addressed in Section 3.5.)*
The third product $\prod_{t\in\bar{\tau}} q_{\sigma,\tau}(\boldsymbol{x}_t \mid \boldsymbol{x}_0)$ looks useless, but it is there because $q_{\sigma,\tau}(\boldsymbol{x}_{1:T} \mid \boldsymbol{x}_0)$ is by definition a density over all T latents and the chain product alone covers only the indices in τ. The simplest valid completion is to set each skipped factor to the DDPM marginal $\mathcal{N}(\sqrt{\alpha_t}\boldsymbol{x}_0, (1-\alpha_t)\boldsymbol{I})$ and to make the skipped variables conditionally independent of each other and of $\boldsymbol{x}_\tau$ given $\boldsymbol{x}_0$. This preserves the DDPM marginal at every t, leaves the chain piece untouched, and keeps the variational objective decomposable into one KL term per timestep. The factorisation is a definition rather than a derivation, and validity is mechanical: each factor is a normalised Gaussian, the integral is one, and the marginals match by an analogue of Lemma 1.

Q4. **While in normal sampling in DDIMs, we had T σ's and T γ's; for accelerated sampling, we have S σ's and T γ's. How does the correspondence work?** *(Addressed in Section 3.5.)*
There are still T σ's in the accelerated case, S chain ones $\sigma_{\tau_i}^2$ from the variances of the chain conditionals (visible in the sampling rule) and $T-S$ star ones σ_t^2 from the variances of the star factors $p_\theta^{(t)}(\boldsymbol{x}_0 \mid \boldsymbol{x}_t)$ for $t \in \bar{\tau}$ (silent at sample time but present in the variational objective). Each one produces a positive γ_t via the same algebra as in the proof of Theorem 1: chain indices give $\gamma_{\tau_i} = c_{\tau_i}^2(1-\alpha_{\tau_i})/(2\sigma_{\tau_i}^2\,\alpha_{\tau_i})$, and skipped indices give $\gamma_t = (1-\alpha_t)/(2\sigma_t^2\,\alpha_t)$ from the star factor in direct analogy with the $t=1$ term. Together they cover all T time-points.

Q5. **If the time steps in $\bar{\tau}$ are in the expression for the inference factorisation for accelerated sampling, why doesn't the generation process depend on them?** *(Addressed in Section 3.5.)*
Two different objects share the symbol p_θ: a joint distribution and a sampling procedure. The joint $p_\theta(\boldsymbol{x}_{0:T})$ in Eq. (25) has to assign a density to every latent $\boldsymbol{x}_t$ for $t \in \{1,\dots,T\}$, so it includes a factor at every index, both chain and star. The sampling procedure is something we choose to define on top of it: pick $\boldsymbol{x}_T \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$ and then traverse the chain factors $\boldsymbol{x}_T \to \boldsymbol{x}_{\tau_{S-1}} \to \cdots \to \boldsymbol{x}_0$. The star factors $p_\theta^{(t)}(\boldsymbol{x}_0 \mid \boldsymbol{x}_t)$ for $t \in \bar{\tau}$ are not transitions on the trajectory, so the procedure has nothing to do with them; producing a sample at $\boldsymbol{x}_0$ does not require ever instantiating the skipped $\boldsymbol{x}_t$. They survive only in the variational objective $J_{\sigma,\tau}$, where each one becomes a denoising loss at a skipped timestep.

Q6. **Do we need to find the corresponding γ values given $\boldsymbol{\sigma}$?** *(Addressed in Section 3.4 and Appendix A.)*
The closed form derived in the proof of Theorem 1 pins γ down algebraically given $\boldsymbol{\sigma}$. Without weight-sharing, this does not matter, because each L_γ summand is decoupled and any positive scalar leaves its minimiser fixed, so $\arg\min_\theta L_\gamma = \arg\min_\theta L_1$ for every positive γ. With weight-sharing the per-t minimisers are coupled and the optimum does depend on the weighting, so being exactly J_σ-optimal would require training on L_γ with the prescribed weights. Ho et al. [2020] and Song et al. [2020a] train on uniform L_1 instead and lean on the empirical observation that the resulting network is fine across the whole $\boldsymbol{\sigma}$-family, the same empirical compromise discussed in Q2.

Q7. **Is it theoretically proven that without variance ($\sigma = 0$), Theorem 1 holds?** *(Addressed in Section 3.4 and Section 5.)*

The proof of Theorem 1 computes a KL between Gaussians of strictly positive variance, which is undefined as $\sigma_t \to 0$, and Song et al. [2020a, Footnote 5] note explicitly that "this case is not covered in Theorem 1, we can always approximate it by making σ_t very small." Two informal arguments fill the gap. The sampling rule is continuous in $\boldsymbol{\sigma}$ at 0, and for any $\boldsymbol{\sigma}_n \to 0$ with $\boldsymbol{\sigma}_n > 0$ Theorem 1 gives weights $\gamma_n \in \mathbb{R}^T_{>0}$ that blow up but stay positive, so $\arg\min_\theta L_{\gamma_n} = \arg\min_\theta L_1$ throughout the sequence and the L_1-trained network remains optimal in the limit.

Independently, Song et al. [2020a, Section 4.3 and Proposition 1] show that the deterministic DDIM update is the Euler step of the variance-exploding probability-flow ODE of Song et al. [2020b], and this interpretation gives a separate justification for the optimality of the L_1-trained network at $\boldsymbol{\sigma} = 0$ without appealing to a limit. (I do not understand this argument well enough to reproduce it here.)